

# SCm CMB Angular Power Spectrum via Phonon-Modulated Primordial Integration

Daniel T. Murphy

2026-04-15

## PAPER\_1092: SCm CMB Angular Power Spectrum via Phonon-Modulated Primordial Integration

### Abstract

We derive the full SCm CMB angular power spectrum by integrating the phonon-modulated primordial power spectrum over  $k$ -space:

$$C_\ell^{\text{SCm}} = \frac{2}{\pi} \int dk k^2 P_{\text{SCm}}(k) |\Delta_T^{\text{SCm}}(\ell, k)|^2$$

where the SCm-modified primordial spectrum is:

$$P_{\text{SCm}}(k) = A_s \left( \frac{k}{k_{\text{piv}}} \right)^{n_s-1} [1 + \Phi \cdot S_{26} \cdot (2R - 1)]$$

This extends the simplified CMBAngularPowerCalculator (grok\_100\_equations\_module.py) to a full  $k$ -space integration with Sachs-Wolfe, acoustic, and Silk damping transfer functions, all modulated by the SCm phonon linewidth  $\Gamma$ .

### \$\$\$1 SCm-Modified Primordial Spectrum

The standard nearly scale-invariant spectrum  $P(k) = A_s (k/k_{\text{piv}})^{n_s-1}$  is boosted by the SCm buoyancy factor:

$$P_{\text{SCm}}(k) = P(k) \cdot [1 + \Phi_{1.25 \text{ THz}} \cdot S_{26}^{(3)}([\text{SSq}]) \cdot (2R - 1)]$$

with  $A_s = 2.1 \times 10^{-9}$  (Planck 2018),  $n_s = 0.965$ ,  $k_{\text{piv}} = 0.05 \text{ Mpc}^{-1}$ ,  $\Phi_0 = 10^{20}$ ,  $S_{26} \approx 1.86$ .

## §§2 Temperature Transfer Function

$$\Delta_T^{\text{SCm}}(\ell, k) = \Delta_T^{\text{SW}}(\ell, k) + 0.6 \cdot \Delta_T^{\text{acoustic}}(\ell, k)$$

### §§2.1 Sachs-Wolfe Term

$$\Delta_T^{\text{SW}} = \frac{1}{3} \exp\left(-\frac{x^2}{2}\right), \quad x = \frac{k \cdot r_*}{\ell + 0.5}$$

where  $r_* \approx 14,000$  Mpc is the comoving sound horizon at last scattering.

### §§2.2 Acoustic Oscillation Term

$$\Delta_T^{\text{acoustic}} = \cos(x) \cdot \exp\left(-\frac{k^2}{k_D^2}\right)$$

with Silk damping scale  $k_D \approx 0.15 \text{ Mpc}^{-1}$ .

## §§3 Full Power Spectrum Integral

$$C_\ell^{\text{SCm}} = \frac{2}{\pi} \int_{k_{\min}}^{k_{\max}} dk k^2 P_{\text{SCm}}(k) |\Delta_T^{\text{SCm}}(\ell, k)|^2 \cdot (T_0 \times 10^6)^2$$

in units of  $\mu\text{K}^2$ .

## §§4 Acoustic Peak Structure

| Peak | Multipole $\ell$ | $C_\ell^{\text{SCm}} (\mu\text{K}^2)$ | Physical Origin        |
|------|------------------|---------------------------------------|------------------------|
| 1st  | $\sim 220$       | Dominant                              | SW + first compression |
| 2nd  | $\sim 546$       | Subdominant                           | First rarefaction      |
| 3rd  | $\sim 800$       | Tertiary                              | Second compression     |

The SCm boost factor  $1 + \Phi S_{26}(2R - 1)$  uniformly amplifies all peaks, preserving the observed peak ratio structure while providing a physical origin via vacuum phonon resonance.

## §§5 Advantages over Standard Treatment

| Aspect              | Standard $\Lambda$ CDM | SCm Phonon                     |
|---------------------|------------------------|--------------------------------|
| Primordial spectrum | Inflaton slow-roll     | SCm vacuum buoyancy            |
| Free parameters     | $A_s, n_s, r$          | $\Phi_0, \Gamma, [\text{SSq}]$ |
| Physical mechanism  | Quantum fluctuations   | Phonon resonance at 1.25 THz   |
| Peak amplification  | Static $A_s$           | Dynamic $\Phi(\Gamma)$         |
| Testability         | CMB polarization       | THz phonon linewidth           |

## References

- PAPER\_1093: SCm CMB Temperature Fluctuation
- PAPER\_1094: CMB Buoyancy Sector Lagrangian
- PAPER\_202: UQFF Reionization / BBN / Recombination
- PAPER\_199:  $F_{U,Bi,i}$  Taxonomy Part 2 — Cosmological Dark Sector

## Key References with arXiv/DOI Identifiers

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